PROJECT SUBMISSION

# Nand to Tetris

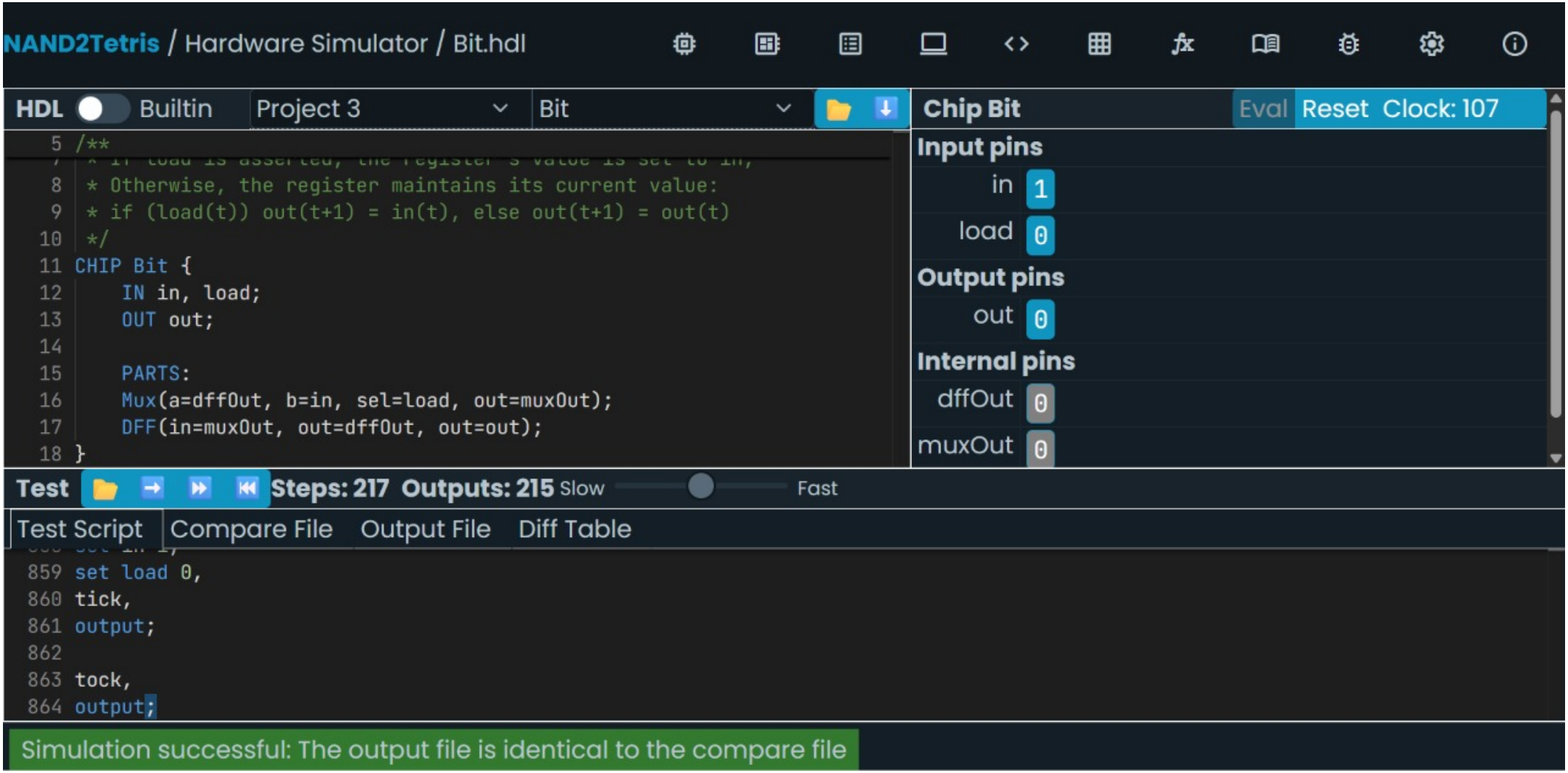
Hardware Simulator Screenshots

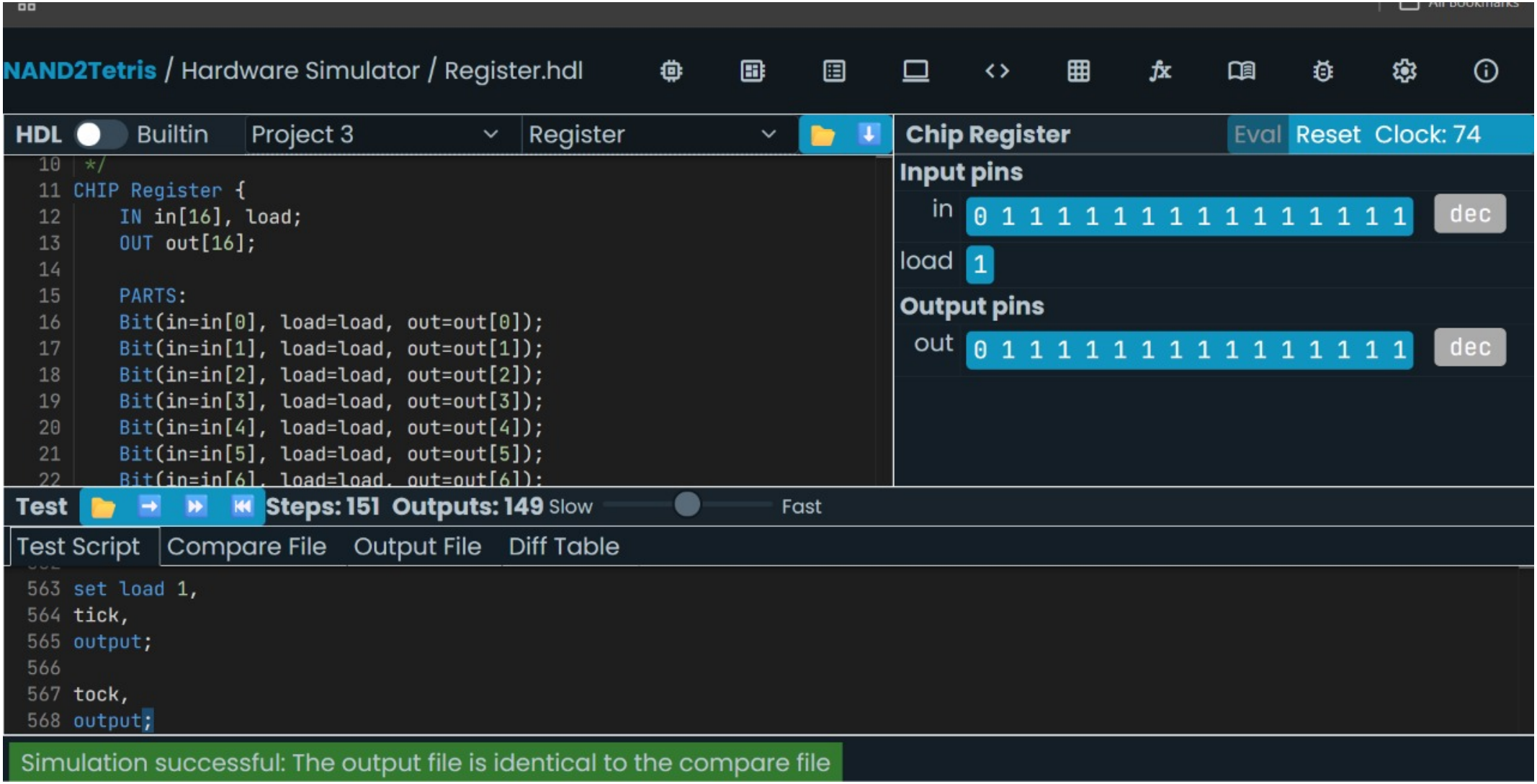

NAME

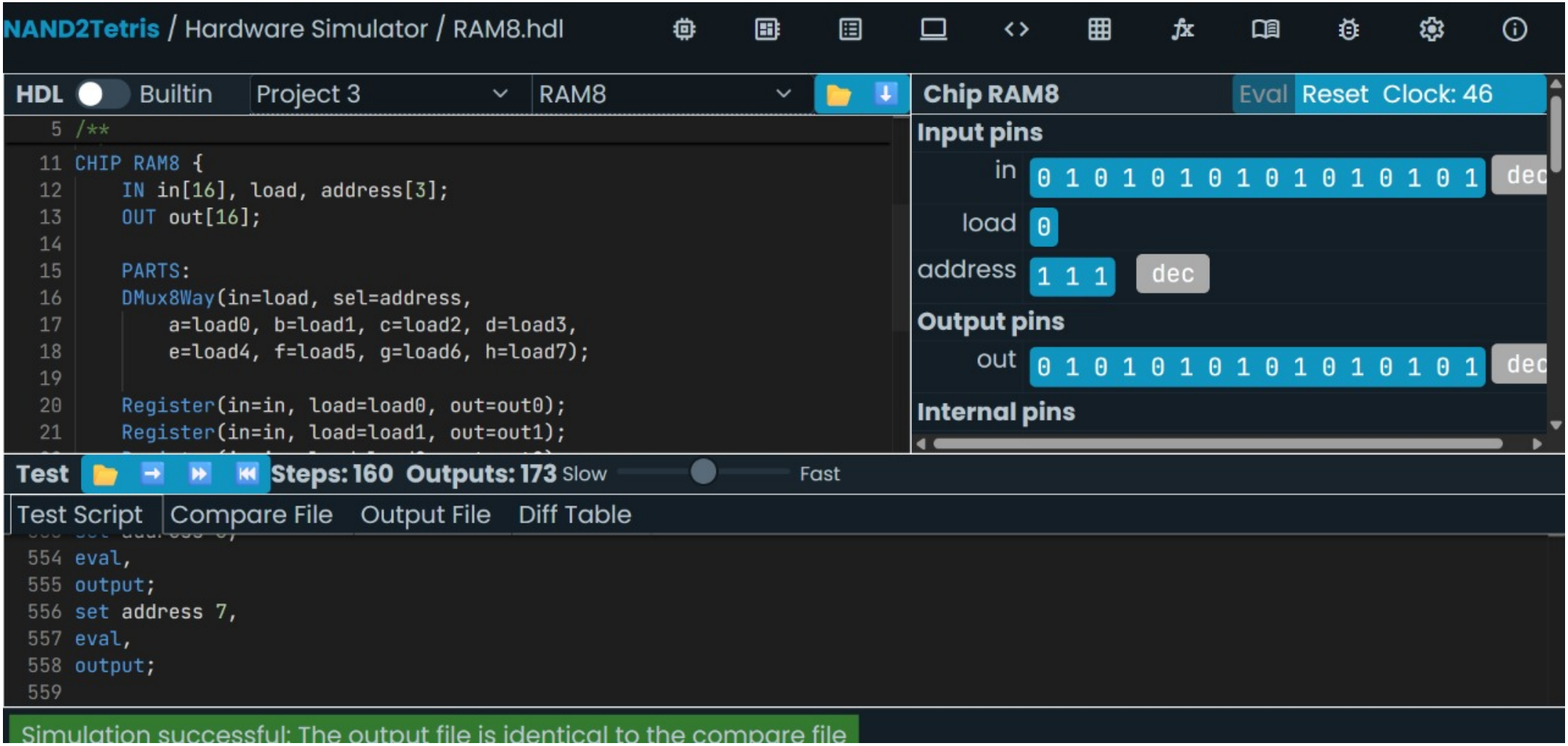
**Prachi**

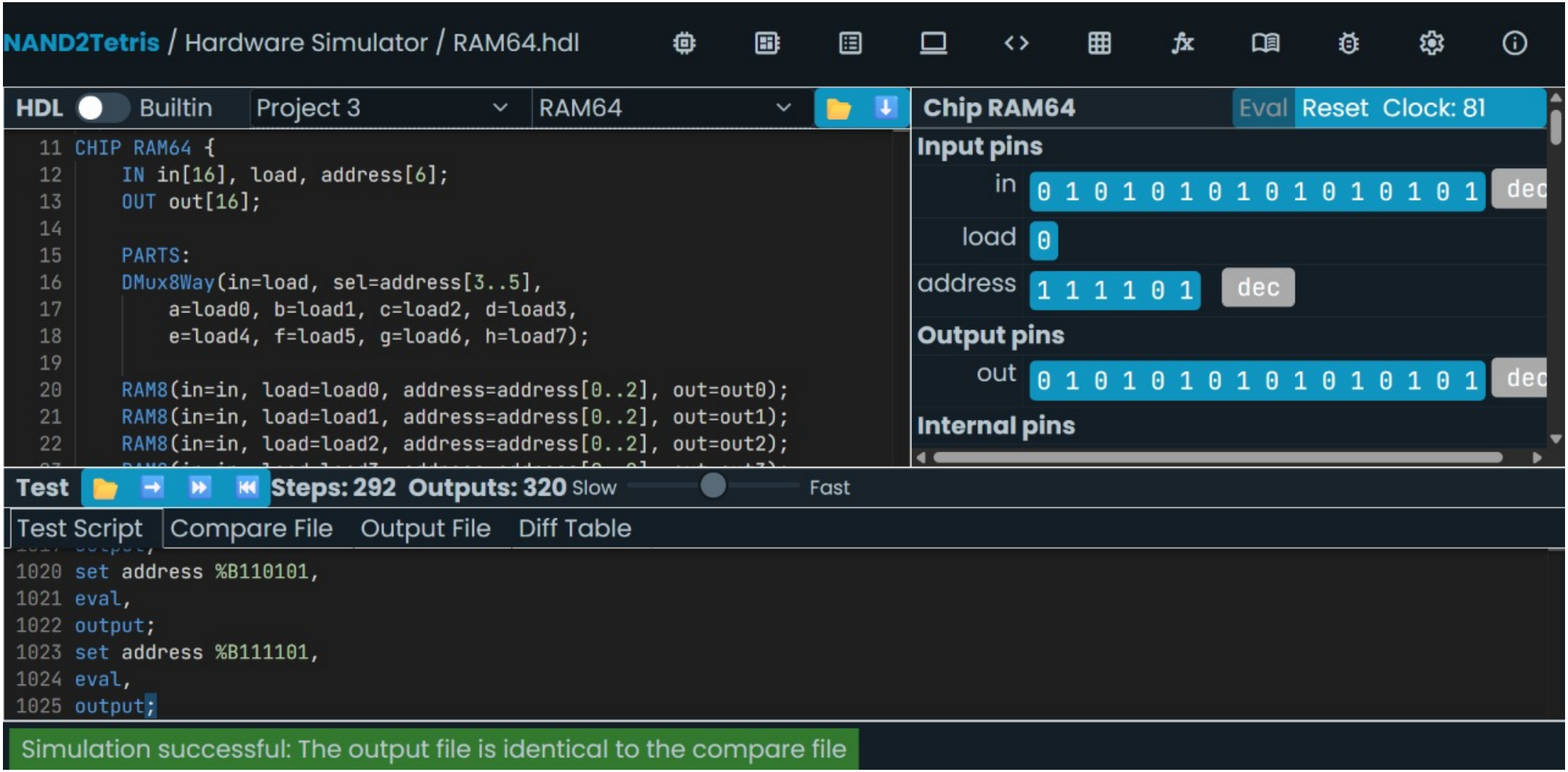
ROLL NO

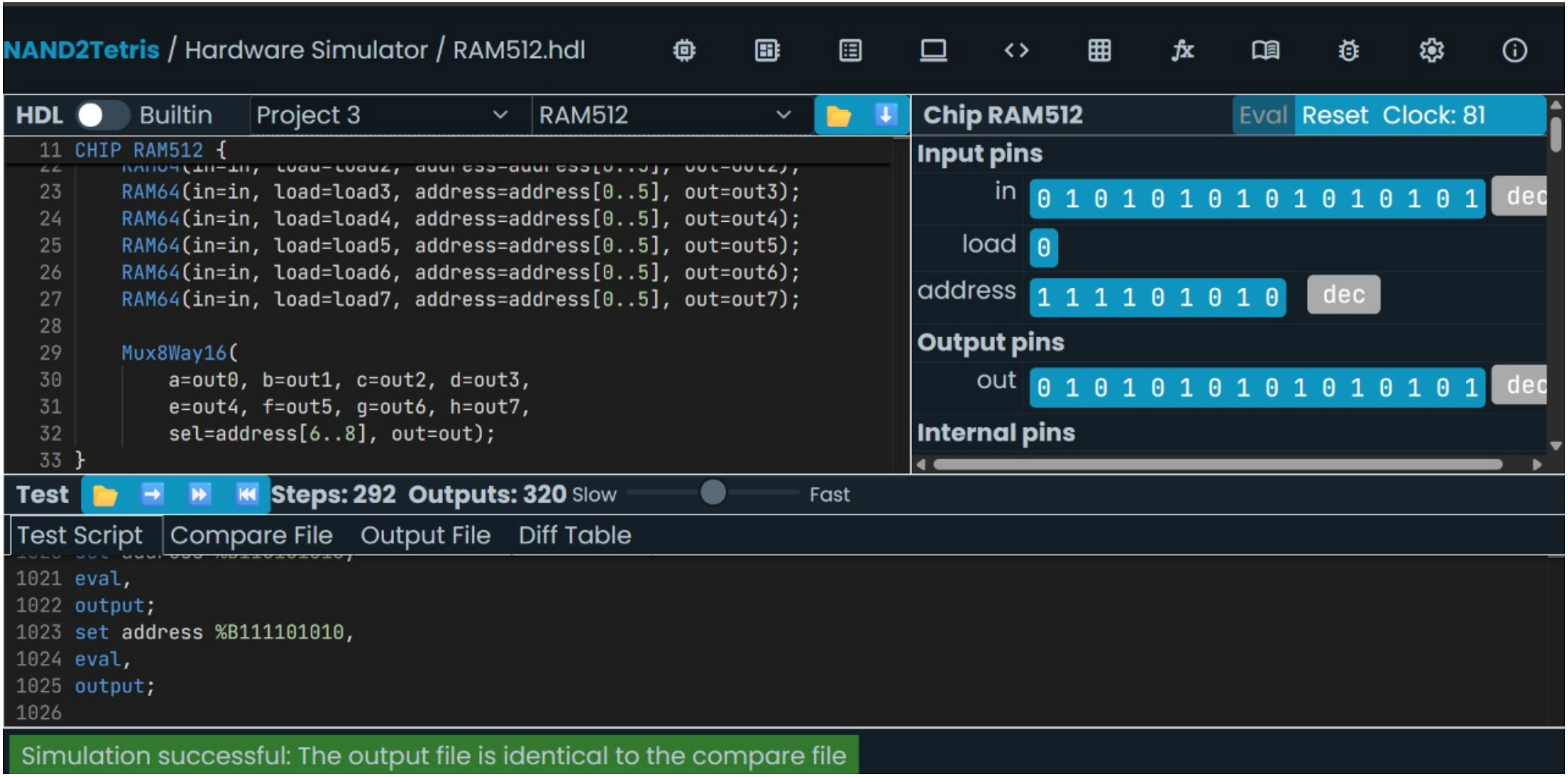
**2501730084**

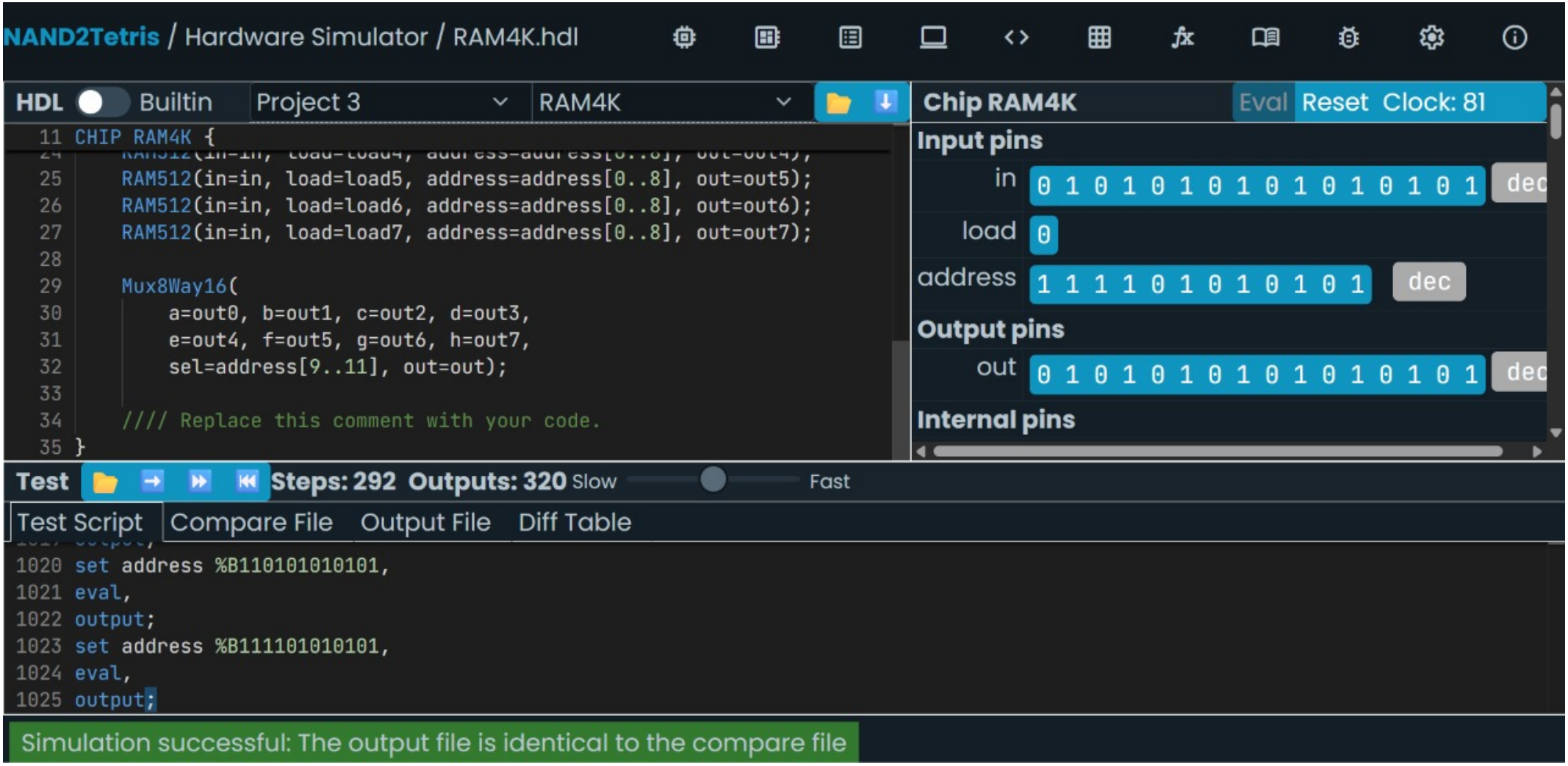

NAND2Tetris / Hardware Simulator / Bit.hdl
HDL
Builtin
Project 3
Bit
5 /**
8 * Otherwise, the register maintains its current value:
9 * if (load(t)) out(t+1) = in(t), else out(t+1) = out(t)
10 */
11 CHIP Bit {
12 IN in, load;
13 OUT out;
15 PARTS:
16 Mux(a=dffOut, b=in, sel=load, out=muxOut);
17 DFF(in=muxOut, out=dffOut, out=out);
18 }
Chip Bit
Eval
Reset
Clock: 107
Input pins
in 1
load 0
Output pins
out 0
Internal pins
dffOut 0
muxOut 0
Test
Steps: 217 Outputs: 215
Slow
Fast
Test Script
Compare File
Output File
Diff Table
859 set load 0,
860 tick,
861 output;
863 tock,
864 output;
Simulation successful: The output file is identical to the compare file

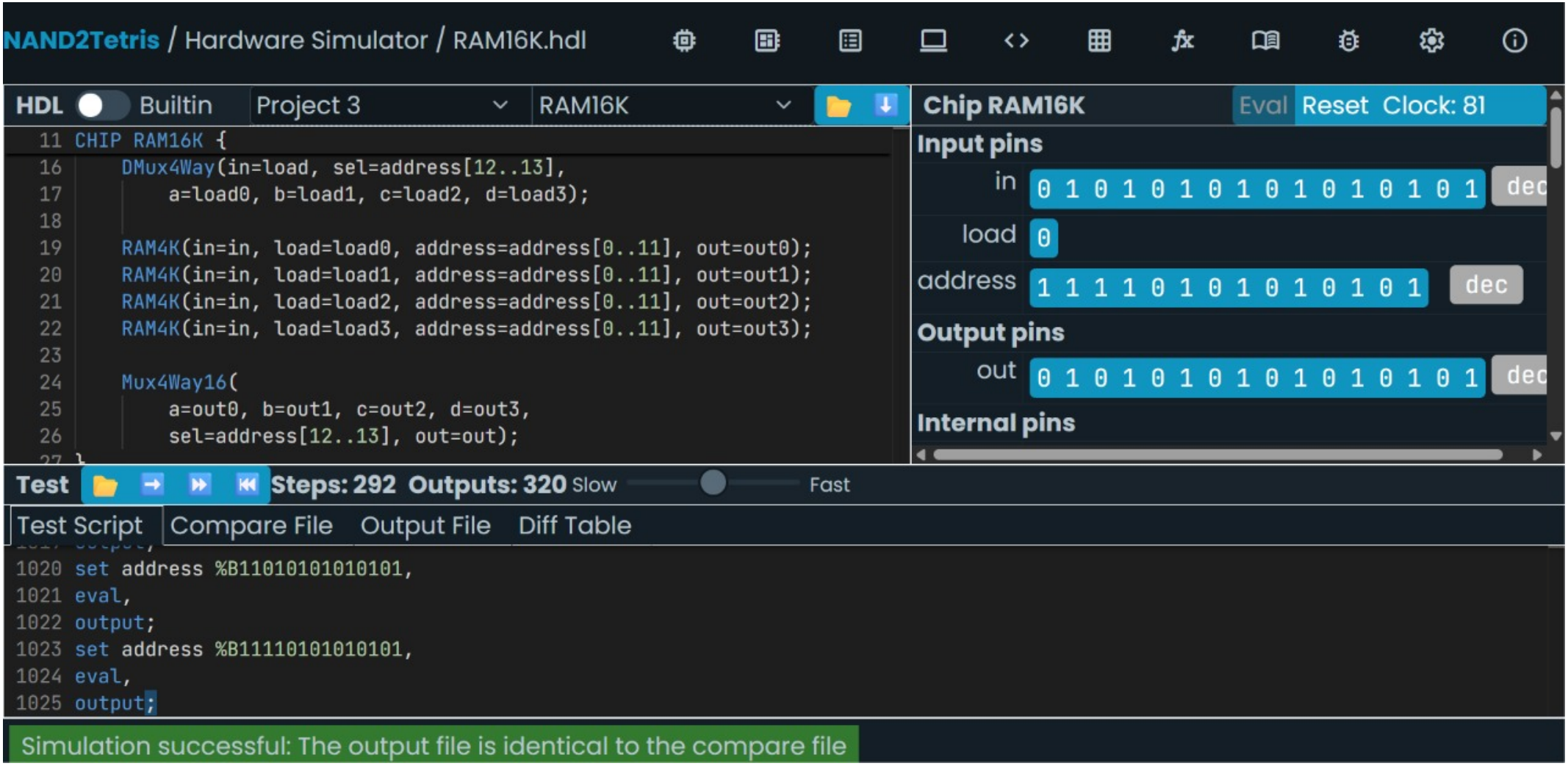

NAND2Tetris / Hardware Simulator / Register.hdl
HDL Builtin Project 3 Register
10 */
11 CHIP Register {
12 IN in[16], load;
13 OUT out[16];
14
15 PARTS:
16 Bit(in=in[0], load=load, out=out[0]);
17 Bit(in=in[1], load=load, out=out[1]);
18 Bit(in=in[2], load=load, out=out[2]);
19 Bit(in=in[3], load=load, out=out[3]);
20 Bit(in=in[4], load=load, out=out[4]);
21 Bit(in=in[5], load=load, out=out[5]);
22 Bit(in=in[6], load=load, out=out[6]);
Chip Register Eval Reset Clock: 74
Input pins
in 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 dec
load 1
Output pins
out 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 dec
Test Steps: 151 Outputs: 149 Slow Fast
Test Script Compare File Output File Diff Table
563 set load 1,
564 tick,
565 output;
566
567 tock,
568 output;
Simulation successful: The output file is identical to the compare file

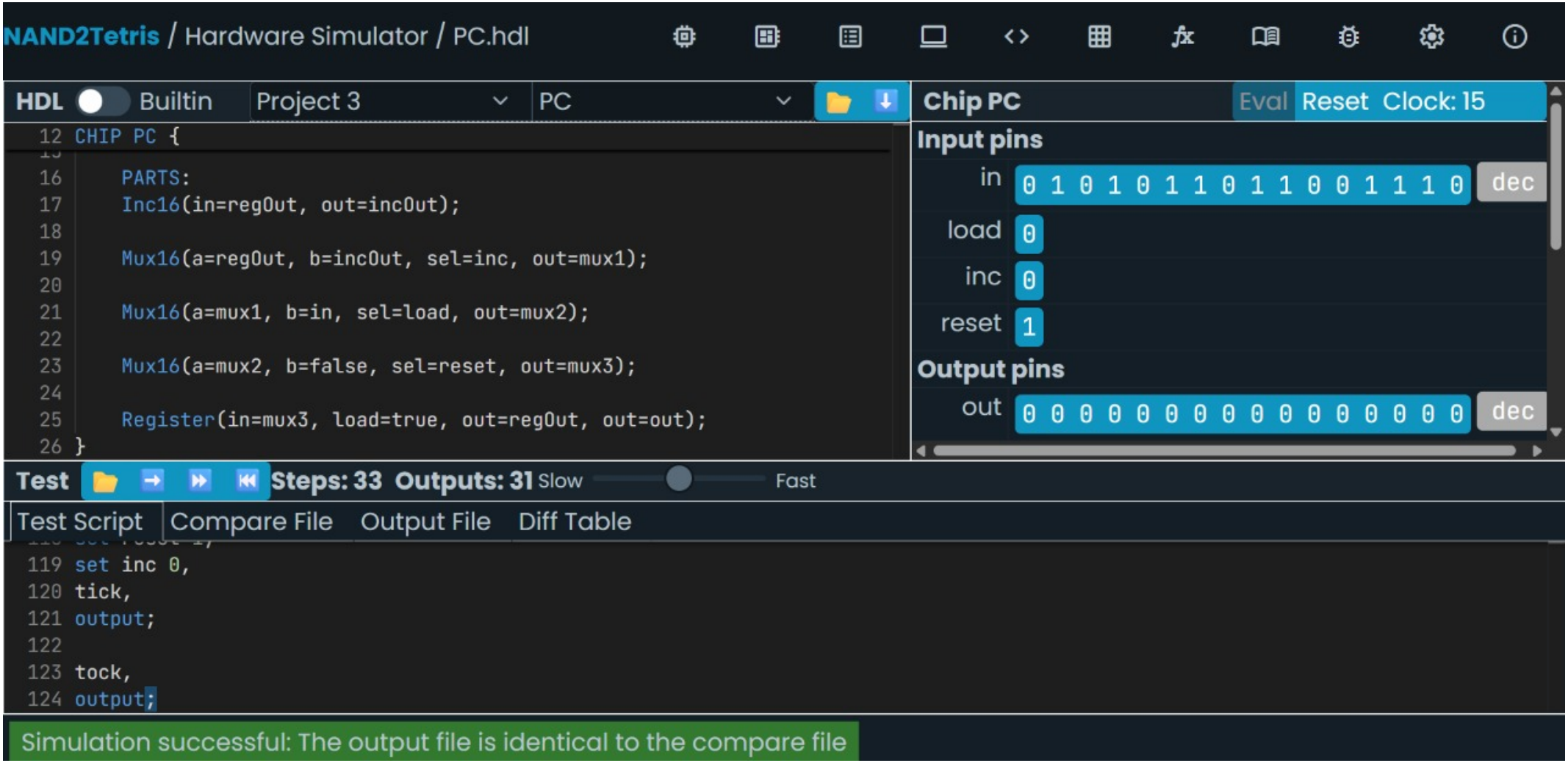

NAND2Tetris / Hardware Simulator / RAM8.hdl
HDL
Builtin
Project 3
RAM8
5 /**
11 CHIP RAM8 {
12 IN in[16], load, address[3];
13 OUT out[16];
15 PARTS:
16 DMux8Way(in=load, sel=address,
17 a=load0, b=load1, c=load2, d=load3,
18 e=load4, f=load5, g=load6, h=load7);
20 Register(in=in, load=load0, out=out0);
21 Register(in=in, load=load1, out=out1);
Chip RAM8
Eval
Reset
Clock: 46
Input pins
in
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
dec
load
0
address
1 1 1
dec
Output pins
out
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
dec
Internal pins
Test
Steps: 160 Outputs: 173
Slow
Fast
Test Script
Compare File
Output File
Diff Table
554 eval,
555 output;
556 set address 7,
557 eval,
558 output;
559
Simulation successful: The output file is identical to the compare file


NAND2Tetris / Hardware Simulator / RAM64.hdl
HDL
Builtin
Project 3
RAM64
11 CHIP RAM64 {
12 IN in[16], load, address[6];
13 OUT out[16];
14
15 PARTS:
16 DMux8Way(in=load, sel=address[3..5],
17 a=load0, b=load1, c=load2, d=load3,
18 e=load4, f=load5, g=load6, h=load7);
19
20 RAM8(in=in, load=load0, address=address[0..2], out=out0);
21 RAM8(in=in, load=load1, address=address[0..2], out=out1);
22 RAM8(in=in, load=load2, address=address[0..2], out=out2);
Chip RAM64
Eval
Reset
Clock: 81
Input pins
in
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
dec
load
0
address
1 1 1 1 0 1
dec
Output pins
out
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
dec
Internal pins
Test
Steps: 292 Outputs: 320
Slow
Fast
Test Script
Compare File
Output File
Diff Table
1020 set address %B110101,
1021 eval,
1022 output;
1023 set address %B111101,
1024 eval,
1025 output;
Simulation successful: The output file is identical to the compare file


NAND2Tetris / Hardware Simulator / RAM512.hdl
HDL
Builtin
Project 3
RAM512
11 CHIP RAM512 {
23 RAM64(in=in, load=load3, address=address[0..5], out=out3);
24 RAM64(in=in, load=load4, address=address[0..5], out=out4);
25 RAM64(in=in, load=load5, address=address[0..5], out=out5);
26 RAM64(in=in, load=load6, address=address[0..5], out=out6);
27 RAM64(in=in, load=load7, address=address[0..5], out=out7);
29 Mux8Way16(
30 a=out0, b=out1, c=out2, d=out3,
31 e=out4, f=out5, g=out6, h=out7,
32 sel=address[6..8], out=out);
33 }
Chip RAM512
Eval
Reset
Clock: 81
Input pins
in
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
dec
load
0
address
1 1 1 1 0 1 0 1 0
dec
Output pins
out
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
dec
Internal pins
Test
Steps: 292 Outputs: 320
Slow
Fast
Test Script
Compare File
Output File
Diff Table
1021 eval,
1022 output;
1023 set address %B111101010,
1024 eval,
1025 output;
1026
Simulation successful: The output file is identical to the compare file


NAND2Tetris / Hardware Simulator / RAM4K.hdl
HDL
Builtin
Project 3
RAM4K
11 CHIP RAM4K {
25 RAM512(in=in, load=load5, address=address[0..8], out=out5);
26 RAM512(in=in, load=load6, address=address[0..8], out=out6);
27 RAM512(in=in, load=load7, address=address[0..8], out=out7);
29 Mux8Way16(
30 a=out0, b=out1, c=out2, d=out3,
31 e=out4, f=out5, g=out6, h=out7,
32 sel=address[9..11], out=out);
34 //// Replace this comment with your code.
35 }
Chip RAM4K
Eval
Reset
Clock: 81
Input pins
in
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
dec
load
0
address
1 1 1 1 0 1 0 1 0 1 0 1
dec
Output pins
out
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
dec
Internal pins
Test
Steps: 292 Outputs: 320
Slow
Fast
Test Script
Compare File
Output File
Diff Table
1020 set address %B110101010101,
1021 eval,
1022 output;
1023 set address %B111101010101,
1024 eval,
1025 output;
Simulation successful: The output file is identical to the compare file


NAND2Tetris / Hardware Simulator / RAM16K.hdl
HDL
Builtin
Project 3
RAM16K
11 CHIP RAM16K {
16 DMux4Way(in=load, sel=address[12..13],
17 a=load0, b=load1, c=load2, d=load3);
18
19 RAM4K(in=in, load=load0, address=address[0..11], out=out0);
20 RAM4K(in=in, load=load1, address=address[0..11], out=out1);
21 RAM4K(in=in, load=load2, address=address[0..11], out=out2);
22 RAM4K(in=in, load=load3, address=address[0..11], out=out3);
23
24 Mux4Way16(
25 a=out0, b=out1, c=out2, d=out3,
26 sel=address[12..13], out=out);
Chip RAM16K
Eval
Reset
Clock: 81
Input pins
in
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
dec
load
0
address
1 1 1 1 0 1 0 1 0 1 0 1 0 1 0 1
dec
Output pins
out
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
dec
Internal pins
Test
Steps: 292 Outputs: 320
Slow
Fast
Test Script
Compare File
Output File
Diff Table
1020 set address %B11010101010101,
1021 eval,
1022 output;
1023 set address %B11110101010101,
1024 eval,
1025 output;
Simulation successful: The output file is identical to the compare file


NAND2Tetris / Hardware Simulator / PC.hdl
HDL
Builtin
Project 3
PC
12 CHIP PC {
16 PARTS:
17 Inc16(in=regOut, out=incOut);
18
19 Mux16(a=regOut, b=incOut, sel=inc, out=mux1);
20
21 Mux16(a=mux1, b=in, sel=load, out=mux2);
22
23 Mux16(a=mux2, b=false, sel=reset, out=mux3);
24
25 Register(in=mux3, load=true, out=regOut, out=out);
26 }
Chip PC
Eval
Reset
Clock: 15
Input pins
in
0 1 0 1 0 1 1 0 1 1 0 0 1 1 1 0
dec
load
0
inc
0
reset
1
Output pins
out
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
dec
Test
Steps: 33 Outputs: 31
Slow
Fast
Test Script
Compare File
Output File
Diff Table
119 set inc 0,
120 tick,
121 output;
122
123 tock,
124 output;
Simulation successful: The output file is identical to the compare file